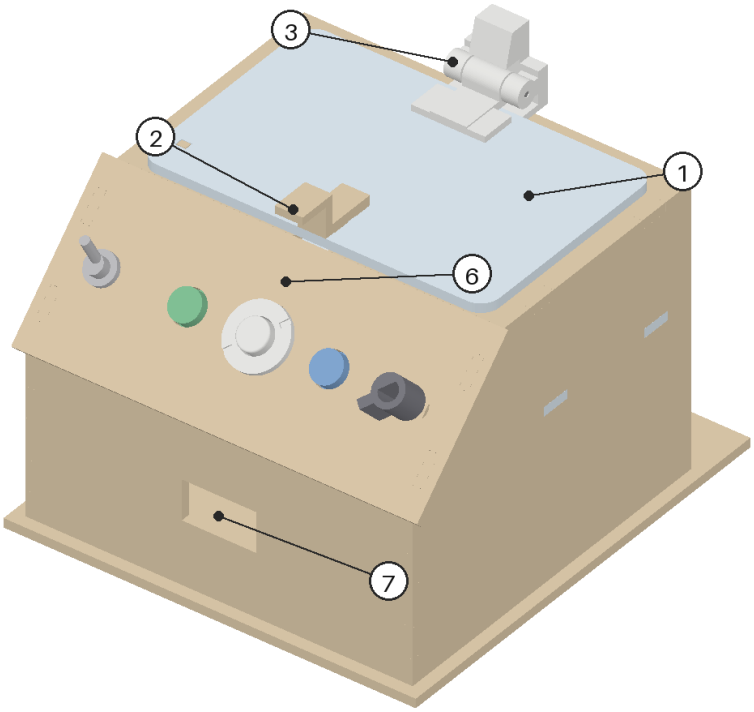
MC INSTRUMENTS

MC-6 Mini Centrifuge

User Manual

Model MC-6 · with MC-R4 tube rack



Read this manual completely before first use. Keep it with the instrument for future reference.

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Quick start

1	Connect the USB cable. Flip **POWER** up to ON. The light ring glows white.
2	Lift the rack cover by its front lip. Lift a tube by its collar or top loop.
3	Lift the centrifuge lid by its front lip. Lower the tube into **well A** until the collar rests on the well.
4	Close the lid. Turn **SPEED** to set the speed; press **TIME+** to add 10 s.
5	Press **START**. Wait for the ring to turn green before opening the lid.

1 About this manual

This manual covers installation, operation, care and troubleshooting of the MC-6 Mini Centrifuge and the MC-R4 tube rack. It is written for laboratory staff and automated handling systems that operate the instrument.

Safety information uses four levels:

Label	Meaning
WARNING	A hazard that may cause serious injury if not avoided.
CAUTION	A hazard that may cause minor or moderate injury.
NOTICE	A situation that may damage the instrument or your samples.
Note	Useful information for correct operation.

2 Safety

Intended use

The MC-6 is a benchtop mini centrifuge for brief spins of MC-6 sample tubes in research and teaching laboratories. Use it only as described in this manual. Do not use it for clinical diagnosis, flammable or explosive samples, or tubes from other manufacturers.

WARNING	Do not open the lid while the light ring shows a run. Press STOP and wait until the ring turns green.
WARNING	When running two tubes, load wells A and B with tubes of the same type and the same fill level. An unbalanced load can make the unit move on the bench.
CAUTION	Pinch point: keep fingers clear of the hinge at the back of the lid and the rack cover when closing them.
CAUTION	Do not move, lift or bump the instrument during a run. Clamp it to the bench using the rear base lip if it will be operated by a robot or used near the bench edge.
NOTICE	Keep liquids away from the housing and the control panel. Wipe spills immediately with the power switched off.
NOTICE	Use only MC-6 tubes, MC-R4 racks and MC Instruments replacement parts. Other tubes may not seat in the wells.

General rules

- Use indoors only, on a firm, level surface, away from heat sources and direct sunlight.
- Power the unit only from a 5 V USB port or adapter rated at 500 mA or more.
- Do not use the instrument if the housing, lid, hinge or cable is damaged.
- Fill tubes no more than two-thirds full, and never force a tube into a well.
- Do not open the housing. There are no user-serviceable parts inside.

3 Product description

The MC-6 has a clear hinged lid over a two-position rotor (wells A and B), a sloped control panel facing the operator, a display on the front, and a light ring under the rotor that shows the run status. Tubes are stored in the MC-R4 rack.

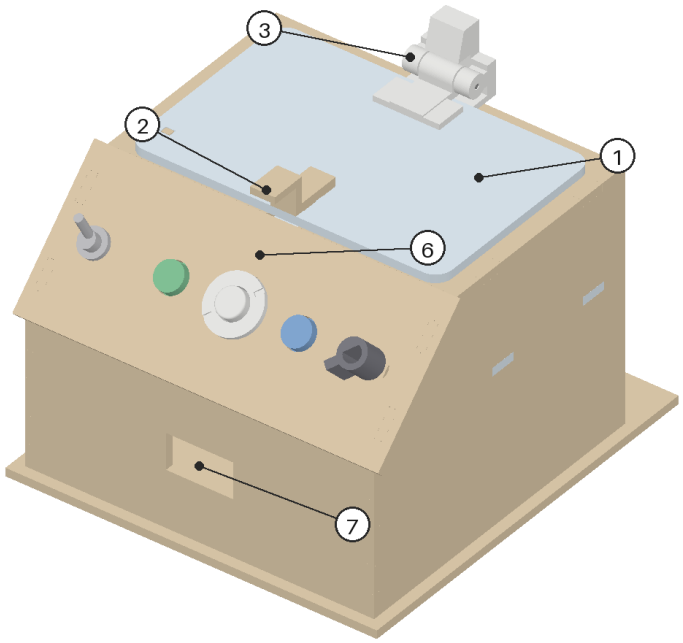


Figure 1. MC-6, lid closed.

No.	Part	Function
1	Lid (clear)	Covers the rotor. Closes under its own weight; opens to about 105° and rests open on its stop.
2	Lid lip	Lift here to open the lid. Designed for one-handed use, including with gloves.
3	Lid hinge	Holds the lid and limits its opening angle. Keep fingers clear.
6	Control panel	POWER, START, STOP, TIME+ and SPEED (see Figure 3).
7	Display	Shows speed, remaining time and messages (section 6).
8	Base lip (rear)	Clamping point for fixing the unit to the bench.

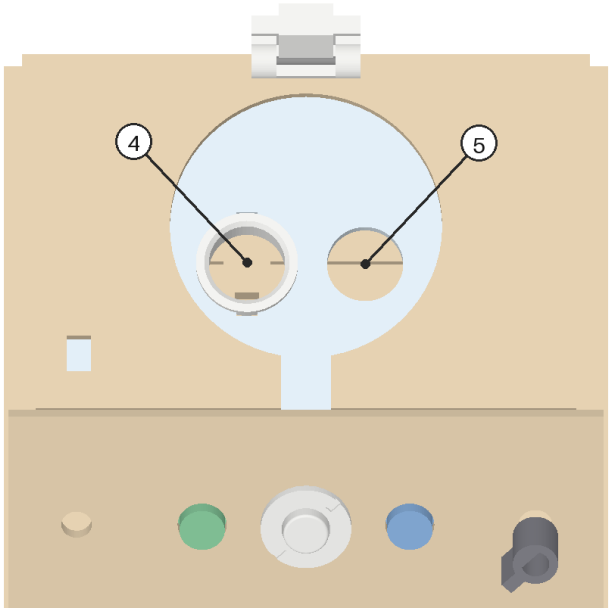


Figure 2. Rotor seen from above (lid removed for clarity).

No.	Part	Function
4	Well A	Main sample position. The run starts only when a tube is fully seated here.
5	Well B	Second position, for a second sample or a balance tube.

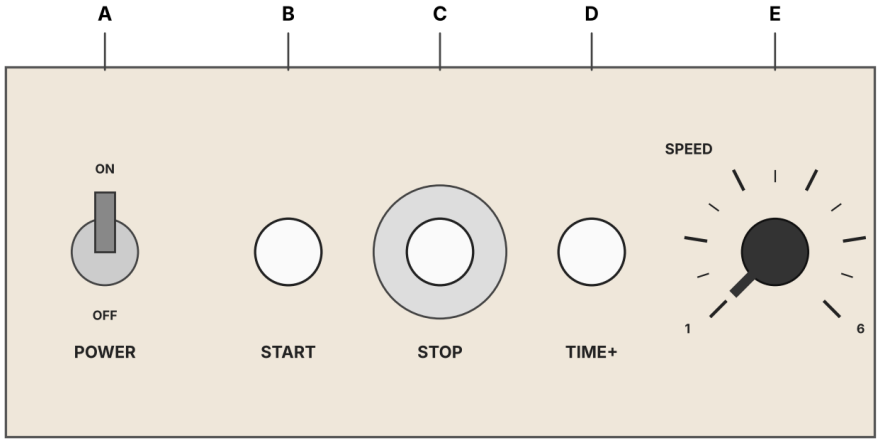


Figure 3. Control panel.

Key	Control	Use
A	POWER switch	Up = ON, down = OFF.
B	START	Starts a run with the speed and time shown.
C	STOP	Ends a run at any time.
D	TIME+	Adds 10 seconds to the run time per press.
E	SPEED dial	Turn clockwise to raise the speed: 1 = 1000 rpm, 6 = 6000 rpm.

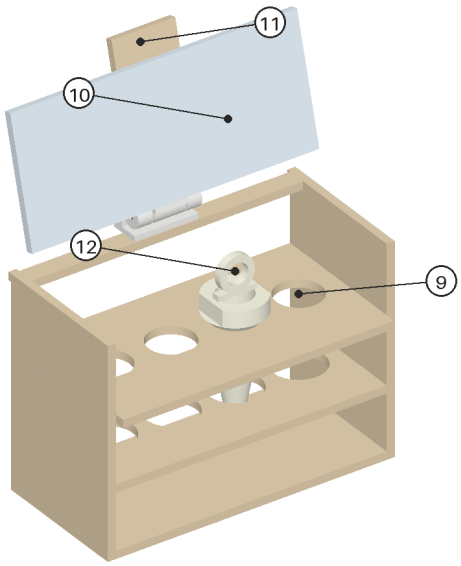


Figure 4. MC-R4 tube rack, cover open.

No.	Part	Function
9	Tube position	Four positions. Lower the tube until its collar rests on the plate.
10	Rack cover (clear)	Protects stored tubes. Opens to about 105° and rests open.
11	Cover lip	Lift here to open the cover.
12	MC-6 tube	Grip the collar flats, or lift by the loop on top.

4 Installation

1. Unpack the centrifuge, the MC-R4 rack, the tubes and the USB cable. Check for transport damage.
2. Place the centrifuge on a level bench with the control panel facing the operator. Leave at least 10 cm free behind it so the lid can open fully.
3. For robot operation, clamp the rear base lip (Figure 1, item 8) to the bench and place the rack within reach, facing the operator position.
4. Connect the USB cable to the socket at the back and to a 5 V USB port.
5. Flip POWER up. The display shows the set speed and time, and the light ring glows white.

Note The settings return to 1000 rpm and 30 s each time the unit is switched on.

5 Operation

5.1 Opening the rack and the lid

1. Lift the rack cover by its front lip until it rests open.
2. Lift the centrifuge lid by its front lip until it rests open on its stop, past vertical.

5.2 Loading a tube

1. Take a tube from the rack by its collar flats or its top loop.
2. Lower the tube straight down into **well A**. Its collar rests on the top of the well when it is seated.
3. For two samples, load the second tube into **well B**, matched in type and fill level.
4. Lower the lid. It closes under its own weight.

NOTICE	Do not press a tube into a well. A correctly sized tube drops in by its own weight. If it sits high, remove it and see section 7.

5.3 Setting speed and time

1. Turn **SPEED** clockwise to raise the speed or counter-clockwise to lower it. The display shows the setting in steps of 100 rpm.
2. Press **TIME+** to add 10 s per press, up to 10 min.

5.4 Running

1. Press **START**. The light ring shows the rotor turning and the display counts down.
2. To end early, press **STOP**.
3. When the time runs out the ring turns green and the display shows **DONE**.
4. Open the lid and lift the tube out by its collar or loop. Return it to the rack and close the cover.

WARNING	If START is refused, the ring turns red and the display shows the reason. Correct it before pressing START again (section 6).

6 Display messages and light ring

Display	Meaning	What to do
1000 RPM 0:30	Ready. Set speed and time.	Load, close the lid, press START.
4500 RPM 0:12	Run in progress, time remaining.	Wait, or press STOP.
DONE	Run complete.	Open the lid and unload.
LID	The lid is open or not fully closed.	Lower the lid fully, then press START.
TUBE	No tube is seated in well A.	Seat a tube in well A so its collar rests on the well, then press START.

Light ring	Status
Off	Power off.
White, steady	Ready.
Blue, rotating	Run in progress.
Green, steady	Run complete. The lid may be opened.
Red, steady	START refused; see the display.

7 Troubleshooting

Problem	Possible cause	Remedy
Display and ring stay dark	POWER is off, or no USB power.	Flip POWER up. Check the cable and use a 5 V port rated 500 mA or more.
Display shows LID after START	The lid is not fully down.	Lower the lid until it rests flat, then press START.
Display shows TUBE after START	No tube in well A, or the tube is sitting high.	Lift the tube and lower it again so the collar rests on the well. Use well A for a single tube.
Tube will not drop into a well	Wrong tube, debris in the well, or a worn well insert.	Use MC-6 tubes only. Check the well for debris. Replace the well A insert (section 8).
A button feels stiff or does not respond	Debris or wear in the button module.	Press the centre of the button. If it still sticks, replace the button module (section 8).
Lid or cover will not stay open	It has not been opened fully.	Lift it past vertical until it rests on its stop.
Lid is hard to open	Lid lip damaged or missing.	Replace the lid lip (section 8).
Speed does not change	Dial at the end of its travel.	Turn the dial the other way. The range is 1000 to 6000 rpm.

8 Care and maintenance

Cleaning

- Switch off and disconnect USB before cleaning.
- Wipe the housing, lid, wells and rack with a soft cloth, dry or lightly dampened with 70 % ethanol. Never spray liquid onto the unit.
- Clean the wells and tube positions weekly, and immediately after a spill. Let everything dry before switching on.

User-replaceable parts

The well A insert, the STOP button module and the lid lip can be replaced without tools.

Part	To remove	To fit
Well A insert	Open the lid and remove any tube. Turn the insert 45° clockwise (seen from above) and lift it out.	Drop the new insert in with its tabs in the notches. Turn it 45° anticlockwise (seen from above) to lock.
STOP button module	Turn the button ring 45° clockwise using the slots in its rim, then pull it straight out. The button comes out with it.	Push the new module in with its tabs in the notches. Turn it 45° anticlockwise to lock.
Lid lip	Slide the lip 3 mm toward the front of the lid, then lift it off.	Push its tabs down through the slots in the lid, then slide it 3 mm toward the hinge to lock.

9 Technical data

Item	MC-6
Speed range	1000 – 6000 rpm, in 100 rpm steps
Run time	10 s – 10 min, in 10 s steps (default 30 s)
Capacity	2 positions (wells A and B)
Compatible tubes	MC-6 tube: 16 mm diameter, 55 mm long, with collar and lifting loop
Lid	Clear, gravity-closing; opens to 105° and rests open
Run start conditions	Lid closed and a tube seated in well A
Power	5 V DC via USB, 500 mA maximum
Dimensions (W × D × H)	133 × 144 × 118 mm, including rear base lip and lid hinge
Environment	Indoor use, 10 – 35 °C, 20 – 80 % RH non-condensing

Item	MC-R4 tube rack
Positions	4
Dimensions (W × D × H)	108 × 52 × 81 mm
Cover	Clear, flip-up; opens to 105° and rests open

10 Warranty and service

MC Instruments warrants the MC-6 and MC-R4 against defects in materials and workmanship for 12 months from the date of purchase. The warranty does not cover damage from misuse, unauthorised modification, liquid ingress or use of parts other than MC Instruments parts. During the warranty period a faulty unit will be repaired or replaced after inspection.

For service, decontaminate the instrument and contact your distributor with the model number and a description of the fault.

Demonstration model built for MakeCU 2026. The rotor does not spin; speeds and run times are shown on the display and the light ring only.